AHAMMAD SHAWKI

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EDUCATION

Bangladesh University of Engineering and Technology (BUET) Expected 2028

BSc in Computer Science and Engineering *Dhaka, Bangladesh*

Coursework: Data Structures and Algorithms, Object Oriented Programming, Database, Digital Logic Design, Computer Architecture, Theory of Computation, Software Engineering, Linear Algebra, Probability and Statistics, Advanced Calculus, Discrete Mathematics, Signals and Linear Systems.

EXPERIENCE

President of Web and App Development, NDITC Mar 2022 to Jun 2024

- Led a 30-person engineering team; delivered the rebuilt official website and the UI/UX for the club mobile app.
- Raised over 1,000,000 BDT across 100+ sponsor negotiations, and taught web development to more than 500 students.

Founder and Team Lead, The AS8 Organization Feb 2021 to May 2022

- Built a developer community reaching 20,000+ learners; partnered with Codecademy, Hack Club and DeepLearning.AI, and wrote for freeCodeCamp News.

SELECTED PROJECTS

DeepSIFT MCP, Python, Volatility 3

- 148 forensic MCP tools parse binaries to typed JSON first. 4/4 on published ground truth, zero hallucinations.

CADENCE Python, PyTorch, speech

- Won best overall. Parkinson's from 30 seconds of voice, validated across three language corpora, not one split.

Momentalk SAM 3.1, Llama, TypeScript

- Photos become speech for people who cannot rely on it. SAM 3.1 outlines every object; tapping one speaks.

Shoulder Strands Agents, A2A, Python

- Private per-sibling agents negotiate a care rota. A reason has no field to travel in. Fair division is code.

Cascade Python, AWS, vector search

- Agent memory that expires. Runbooks are pinned to policy versions; one edit invalidates them all in four writes.

TRACE Python, signal processing

- Heart rate from webcam colour with no trained model. 4.62 BPM error against 5.63 for the best published method.

Someone Snap SPECS, Lens Studio, CLAD

- Hands-free CPR on AR glasses: a sternum target at real size and a ring pulsing at 110 a minute.

Locus GitLab Duo agents, Cloud Run

- Agents turn an emailed function into a raised-line braille graph, check it, ship it and order the print.

MACHINE LEARNING COMPETITIONS

Historic handwriting transcription R.O.A.D. Barbados, Zindi

- 0.92385 against a winning 0.93306, from an MBR vote over 34 fine-tuned readers on free Kaggle hardware.

Bengali LLM hallucination detection IUTCS Datathon 2.0

- Third at 0.922 F1. A decision ladder resolves most rows deterministically; only the residual reaches a judge.

DaT Parkinson's Challenge DrivenData

- Tenth on the private leaderboard, 0.2760 log loss and 0.9551 AUROC, from registration plus stacked nets.

Phantom Road Gaps, Bias Bounty Humane Intelligence and Zindi

- Double-counted boundary highways are 72% of the mean road gap, and they bend the equity scorecard.

AWARDS

- **Fifth place, Top 10 finalist**, Data Privacy Avengers 2026, Robi Axiata
- **Winner, Golden Agent**, Agents for Humans Hackathon 2026, for Shoulder
- **Grand Prize Winner**, FLIR App Challenge 2025 to 2026, for SolarSnap
- **Best Overall Project**, ML Empowerment Build Challenge 2.0, for CADENCE
- **Best ERNIE Multimodal Application**, Baidu AI Developer Challenge
- **Honourable Mention**, LMA EDGE Hackathon, London finale, for Coven

TECHNICAL SKILLS

Languages: Python, C/C++, TypeScript, JavaScript, Java, SQL

Machine learning: cross-corpus validation, domain-adversarial training, ensembling and stacking, probability calibration, image registration, classical signal processing, speech recognition

AI systems: multi-agent orchestration, tool use and the Model Context Protocol, retrieval augmented generation, per-claim grounding, hallucination detection, evaluation harness design, privacy boundaries enforced in types

Tools: PyTorch, React, Next.js, Django, FastAPI, PostgreSQL, Docker, AWS, Google Cloud

Systems: microservice architecture, API design and gateways, relational and vector data modelling, observability and CI/CD, cloud deployment on AWS and GCP, containerisation